UNIVERSITY
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Final report

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Company name: APP MAKERS

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Acknowledgments

Dear App Makers and Esteemed Supervisors,

I hope you are doing well. I am writing to express my heartfelt gratitude for the incredible opportunity to participate in the summer training program at App Makers. This experience has been precious, and I am truly grateful for the chance to learn and grow under your guidance.

Throughout the training, I have gained a deep understanding of the Dart language, which has enabled me to write more efficient and effective code. The sessions on the Flutter framework were particularly enlightening, as they equipped me with the skills to develop sophisticated and responsive mobile applications. The hands-on projects were challenging yet rewarding, allowing me to apply theoretical knowledge in practical scenarios.

The teamwork skills I developed during this period have also been invaluable. Collaborating with my peers on various projects has taught me the importance of communication, coordination, and mutual support in achieving our goals. I now feel more confident in working effectively within a team, and I am excited to bring this collaborative spirit to future projects.

I would like to extend a special thank you to my supervisors, Mahmoud Lotfy and Amal Nassem, for their constant support, encouragement, and expertise. Your feedback and guidance have been instrumental in my learning process. Whether it was through detailed code reviews, insightful discussions, or constructive criticism, each interaction has helped me grow as a developer and a professional.

I am also grateful for the resources and tools provided during the training. Access to platforms like Visual Studio Code and Adobe XD has been crucial in enhancing my technical skills. Learning how to navigate these tools and integrate them into my workflow has been an eye-opening experience.

Once again, thank you for this remarkable opportunity. I look forward to applying what I have learned and contributing positively to our projects and the company. I am proud to be a part of App Makers and am eager to continue my journey here.

Warm regards,
Raghad Mohammed Alghamdi.

Abstract

This report presents a comprehensive overview of the summer training program undertaken at App Makers. The program was designed to enhance participants' proficiency in the Dart language and the Flutter framework, essential for developing efficient and high-quality mobile applications. Throughout the training, participants engaged in a variety of hands-on projects, allowing them to apply theoretical knowledge to practical scenarios and improve their coding skills.

In addition to technical training, the program emphasized the development of teamwork and collaboration skills. Participants worked closely with peers on group projects, fostering a cooperative learning environment and improving their ability to communicate and coordinate effectively within a team.

The guidance and support provided by supervisors, notably Mahmoud Lotfy, were crucial to the participants' success. Their expertise, constructive feedback, and mentorship significantly contributed to the learning experience, helping participants overcome challenges and achieve their goals.

Key achievements of the training include mastering the Dart language, gaining proficiency in the Flutter framework, and completing several mobile application projects. The report also discusses the challenges faced during the training and the strategies employed to address them.

Overall, the summer training program at App Makers has had a profound impact on the participants' professional development, equipping them with the skills and confidence needed to contribute effectively to future projects. This report encapsulates the learning journey, key takeaways, and the overall benefits of the training program.

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Introduction

App Makers is a leading software development company renowned for its innovative and high-quality mobile applications. Founded with a vision to revolutionize the tech industry, App Makers has consistently pushed the boundaries of what's possible in mobile app development. The company prides itself on its commitment to excellence, leveraging cutting-edge technology and user-centric design to create applications that meet and exceed client expectations. With a diverse portfolio spanning various industries, App Makers has established itself as a key player in the tech world, known for its versatility and ability to deliver customized solutions.

The company's culture fosters a collaborative and dynamic work environment where creativity and technical expertise thrive. Employees at App Makers are encouraged to think outside the box, experiment with new ideas, and continuously improve their skills. This nurturing atmosphere was evident during my summer training program, which was meticulously designed to provide interns with a comprehensive learning experience.
The summer training program at App Makers aimed to equip interns with practical skills and in-depth knowledge of the Dart language and the Flutter framework, both essential for modern mobile application development. The program's structure was thoughtfully crafted to balance theoretical learning with hands-on projects, ensuring that interns could apply their newly acquired knowledge in real-world scenarios.

Throughout the training, we were involved in a variety of projects that challenged us to think critically and solve complex problems. These projects ranged from developing simple applications to more sophisticated ones, requiring us to leverage the full spectrum of our coding skills. Additionally, the training emphasized the importance of teamwork and collaboration. Working closely with peers on group projects not only improved our ability to communicate and coordinate effectively but also fostered a sense of camaraderie and shared purpose.

Under the guidance of experienced supervisors and mentors, we received invaluable support and feedback. Their expertise and mentorship were crucial in helping us navigate the challenges we encountered and achieve our learning objectives. This report delves into the specifics of these experiences, highlighting key achievements and reflecting on the professional growth attained during the training period.

In summary, the summer training program at App Makers provided a robust foundation in mobile application development, significantly enhancing our technical and teamwork skills. This introduction sets the stage for a detailed exploration of the training program's components, objectives, and outcomes, showcasing the profound impact it has had on my professional development.

Training Plan

Week 1:

- Project idea selection, team setup, and role delegation.
- Software installation (e.g., Android Studio).
- Initial UI sketches, metadata, and ER diagram structure.
- Regular feedback meetings with the supervisor.

Week 2:

- Learning Adobe XD for UI/UX design.
- Designing primary application interface and ensuring smooth navigation.
- Documenting progress and troubleshooting.

Week 3:

- Finalizing UI/UX design and technical improvements.
- Exporting Adobe XD projects, studying Flutter, and debugging the app.

Week 4:

- Resolving Adobe XD plugin issues, managing library dependencies, and exporting design code.
- Developing a new Flutter page, structuring project files, and implementing a state management pattern.

Week 5:

- Integrating Firebase into the application.
- Researching Firebase services, adding images/icons, and enhancing navigation flow.
- Addressing compatibility issues with Java and Gradle in Flutter and gaining experience in cloud-based solutions.

Week	Work Description	Department	Start Date	Duration (hours)	Training Supervisor
1	Set up Flutter project, configure environment settings, initial project structure		**June 23**	20	
2	Design UI for main screens (home, search, hotel display)		**June 30**	20	
3	Implement home and search screens using Flutter widgets		**July 7**	25	
4	Implement hotel display screen and integrate with dummy data		**July 14**	20	
5	Develop and integrate API calls for hotel search and display		**July 21**	25	
6	Implement and integrate user authentication (login, registration)		**July 28**	20	
7	Implement booking functionality and integrate with backend APIs		**August 4**	25	
8	Test and debug all implemented features, ensure smooth UI/UX		**August 11**	20	
9	Optimize performance, finalize documentation, and prepare for deployment		**August 18**	25	
			Total of hours	200	

Figure 1.Training Plan

Task Description: First Week

Day 1: Initial Setup and Planning

Meeting with the Training Supervisor The week began with a crucial meeting involving all project members and the training supervisor. This initial meeting set the foundation for our project, providing an opportunity to discuss the overall objectives and expectations. The supervisor outlined the key milestones and deliverables, ensuring that everyone had a clear understanding of the project's scope.

Discussion on Member Roles Following the meeting, we deliberated on how to effectively divide the project members into teams or roles. This discussion focused on maximizing efficiency by leveraging each person's strengths and skills. We aimed to ensure that all members could contribute meaningfully to the project.

Idea Selection After thorough discussion, we selected a project idea that was deemed appropriate and feasible for all members to work on. This decision involved considering the interests, skills, and availability of each member, ensuring that the chosen project aligned well with our collective capabilities.

Determining Programs and Tools We identified the necessary programs and tools required for the project. This step was crucial to ensure everyone was on the same page regarding the technical requirements. We discussed the tools needed for design, development, and collaboration, including Adobe XD, Android Studio, and other relevant software.

Day 2: Project Planning and Initial Design

Writing the General Project Plan The second day was dedicated to drafting a comprehensive general plan for the project. This plan outlined the project's objectives, key milestones, timelines, and deliverables. We aimed to create a clear roadmap that would guide our efforts throughout the development process.

Initial Interface Design on Paper We began the design process by sketching the initial interfaces on paper. This step was crucial for visualizing the user experience and functionality before moving to digital design. We focused on creating intuitive and user-friendly interfaces that would meet the project's requirements.

Supervisor Review Everything drafted and designed was discussed with the supervisor. Their feedback and advice were integral in refining our approach and ensuring alignment with the project goals. The supervisor's insights helped us identify potential improvements and areas that needed more attention.

Day 3: Metadata and Database Design

Team Meeting on Metadata and ER Model We organized a meeting with all project members to ensure a unified understanding of the project's metadata requirements. This meeting was essential for aligning our approach to data management and storage.

Drawing the ER Model, we collaborated on drawing the Entity-Relationship (ER) model, which is essential for database design. This involved identifying the entities, relationships, and attributes necessary

for our project. The ER model provided a clear blueprint for how data would be structured and managed within the application.

Day 4: Finalizing Designs and Preparation

Completion of Interface Design We finalized the paper designs of the project interfaces. These designs were comprehensive and included all the necessary details for a seamless user experience. The finalized designs served as a detailed guide for the digital design phase.

Supervisor Review The completed designs were reviewed by the supervisor. Their insights helped us identify any potential issues and make improvements. This review ensured that our designs were robust and aligned with best practices.

Downloading Android Studio, we ensured that all members downloaded and set up Android Studio, the primary development environment for our project. This step was crucial for preparing the team for the development phase. Having the right tools in place allowed us to seamlessly transition from planning and design to actual development.

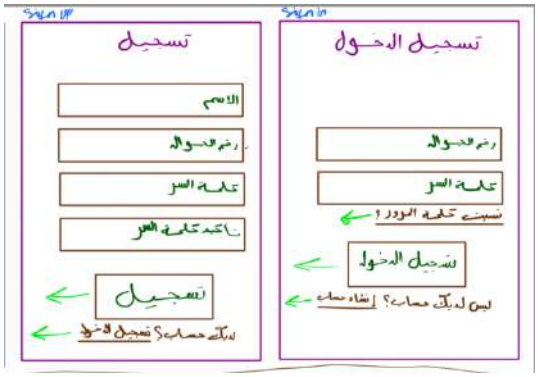


Figure 2.Login and register sketch.

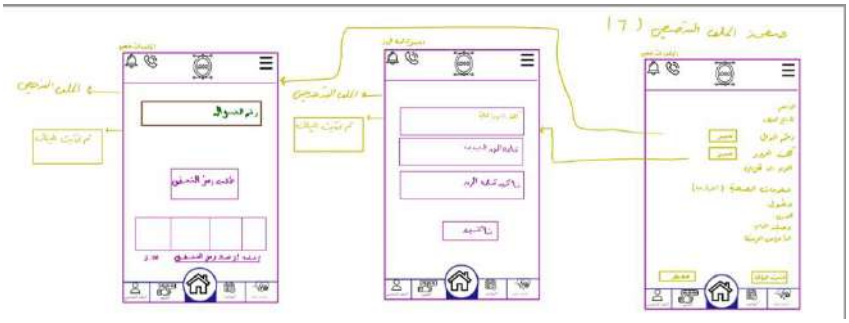


Figure 3.Clinic Profile sketch.

Task Description: Second Week

Day 1: Setting Up Adobe XD

Downloading XD Program, the week began with all team members downloading Adobe XD, the primary tool we chose for designing the final user interfaces. Adobe XD was selected for its robust features and ease of use in creating high-fidelity prototypes. Ensuring every team member had Adobe XD installed and properly configured was crucial for the collaborative design process ahead.

Day 2: Initial Digital Design

Initial Interface Design in XD We began the detailed design process by creating the first set of interfaces in Adobe XD. These digital designs were more refined compared to the initial paper sketches from the previous week. We incorporated feedback from the first week's discussions to enhance the user experience and functionality of the interfaces. This phase involved meticulous attention to detail to ensure the designs were visually appealing and user-friendly.

Review with Supervisor After designing a portion of the interfaces, we scheduled a review session with our supervisor. This session was crucial for identifying and correcting any errors or inconsistencies in the design. The supervisor’s feedback helped us refine our approach, ensuring the designs met the project’s requirements and adhered to best practices in UI/UX design.

Day 3: Continued Design Development

Continuation of Interface Design We continued working on the interface designs in Adobe XD. Building upon the initial designs, this day was dedicated to incorporating the feedback received from the supervisor. Our focus was on enhancing the functionality and aesthetics of the interfaces, ensuring they provided a seamless and intuitive user experience. Collaborative efforts among team members helped in sharing ideas and improving the overall design quality.

Day 4: Finalizing Designs

Completion of Interface Design By the fourth day, we completed the design of all project interfaces in Adobe XD. This milestone included ensuring all design elements were consistent, user-friendly, and aligned with the project's goals. We conducted thorough checks to verify that the interfaces were coherent and that navigation between screens was intuitive. Final touches were added to polish the designs and ensure they were ready for the next phase of development.

Day 5: Linking Interfaces

Linking Interfaces Together The final task for the week was to link all the designed interfaces together. Using Adobe XD's prototyping features, we created interactive links between the screens to simulate the user flow within the application. This step was essential for visualizing how users would navigate through the app and identifying any potential issues in the user journey. By the end of the day, we had a complete interactive prototype that provided a realistic representation of the final application.

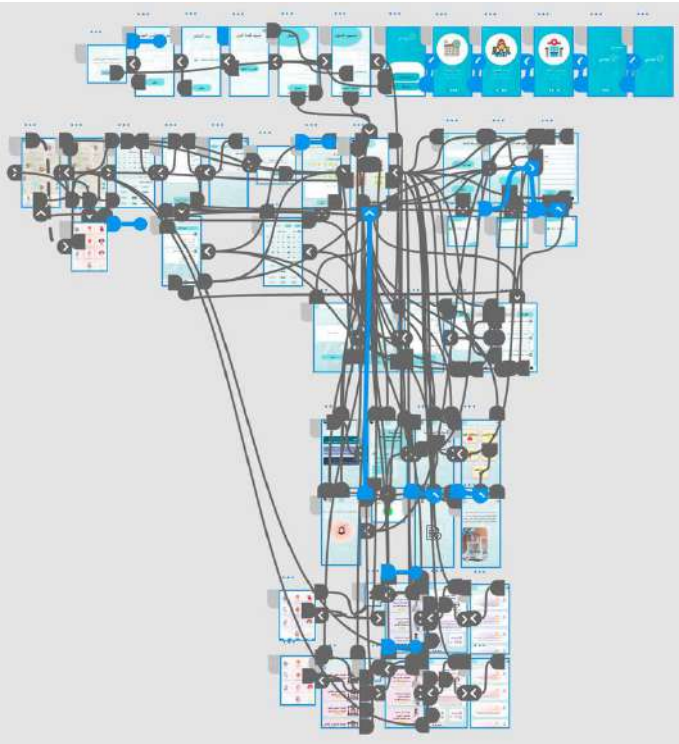


Figure 4.All Interfaces prototypes.

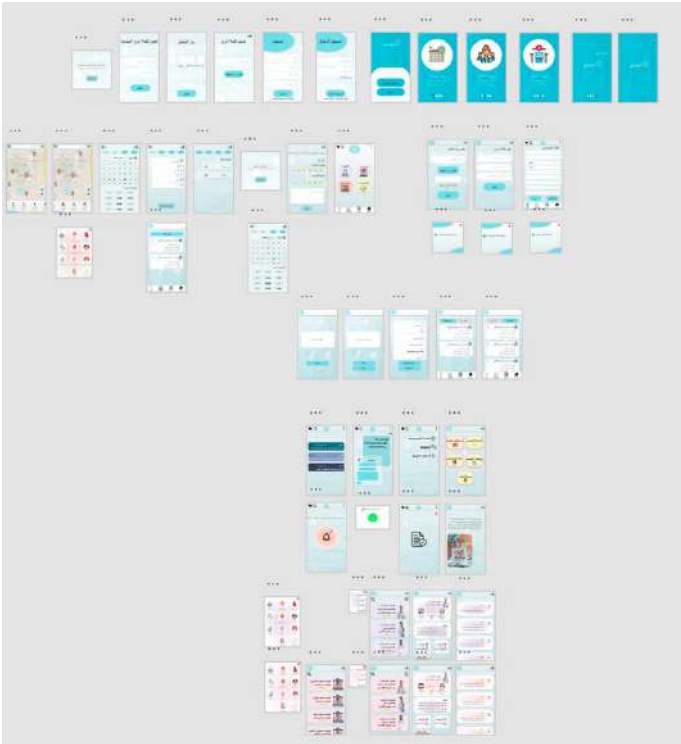


Figure 5.All Interfaces Design.

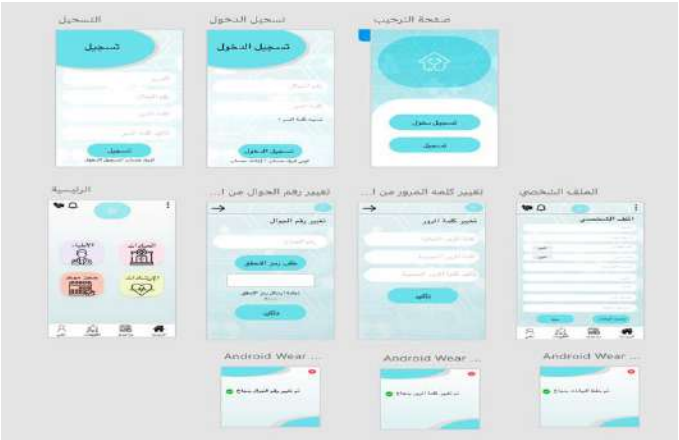


Figure 6.Interface Design.

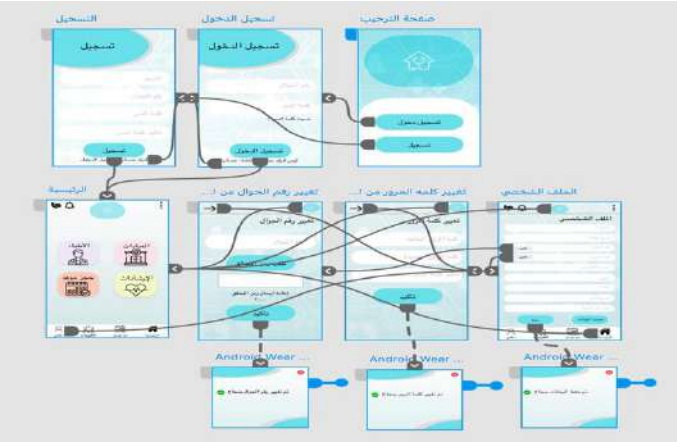


Figure 7.Interface prototype.

Task Description: Third Week

Day 1: Refining Interfaces

Making Modifications to Interfaces We focused on making necessary modifications to the interfaces designed in Adobe XD. This involved refining the user interface to enhance usability and ensure alignment with the overall project requirements. Key activities included:

- **Layout Adjustments**: Fine-tuning the placement of elements to improve visual hierarchy and user flow.
- **Color Scheme Updates**: Adjusting colors to ensure consistency and adherence to design guidelines.
- **Functionality Improvements**: Enhancing interactive elements and ensuring they functioned as intended.

These refinements aimed to create a more polished and user-friendly design, addressing any issues identified during previous reviews.

Day 2: Learning Visual Studio Code

Exploring Visual Studio Code, we dedicated the day to learning about Visual Studio Code (VS Code), a powerful and popular code editor. Our focus included:

- **Navigating the Interface**: Understanding the layout and features of VS Code, including its editor, terminal, and sidebar.
- **Utilizing Extensions**: Exploring useful extensions that enhance productivity and support various programming languages.
- **Debugging Tools**: Learning how to use VS Code's debugging features to identify and fix code issues efficiently.

By the end of the day, we were comfortable setting up our development environment and using VS Code for our programming tasks, laying the groundwork for the next phase of development.

Day 3: Introduction to Dart

Learning the Basics of Dart We began our journey into the Dart programming language using a dedicated book. This introductory session covered:

- **Fundamental Concepts**: Understanding variables, data types, and basic syntax.
- **Control Structures**: Learning about conditionals and loops to control the flow of programs.
- **Basic Syntax**: Familiarizing ourselves with Dart's syntax to write and understand simple code snippets.

This foundational knowledge was crucial for building our understanding of Dart and preparing us for more advanced topics.

Day 4: Advanced Dart Concepts

Deepening Dart Knowledge We continued exploring Dart, delving into more advanced topics. This day was focused on:

- **Object-Oriented Programming (OOP)**: Learning about classes, objects, inheritance, and encapsulation.
- **Asynchronous Programming**: Understanding futures, async/await, and how Dart handles asynchronous operations.
- **Application Development**: Exploring how Dart can be used to build real-world applications and practical use cases.

By the end of the day, we had a comprehensive understanding of Dart's capabilities, equipping us with the knowledge needed for robust application development.

Day 5: Mastering Dart

Solidifying Dart Knowledge To solidify our understanding of Dart, we engaged in:

- **Revisiting Complex Concepts**: Reviewing and deepening our grasp of advanced Dart features and principles.
- **Coding Exercises**: Practicing coding challenges and exercises to reinforce our learning and problem-solving skills.
- **Building Small Projects**: Applying our knowledge by creating small projects, which helped us implement Dart in practical scenarios.

Figure 8.Flutter book.

Task Description: Fourth Week

Day 1: Exporting from Adobe XD to Visual Studio Code

Exporting from Adobe XD On the first day, we focused on exporting our designs from Adobe XD to Visual Studio Code. This step was crucial for transitioning from design to development. However, we encountered issues with the plugins required for this process.

Solving Plugin Issues To complete the export successfully, we identified and resolved problems related to the necessary plugins. This involved:

- **Researching the Plugins**: Understanding which plugins were needed and their specific functionalities.
- **Troubleshooting Errors**: Identifying the issues causing the export process to fail and finding solutions.
- **Installing and Configuring Plugins**: Ensuring the correct installation and configuration of the plugins in Visual Studio Code.

By overcoming these challenges, we ensured a smooth transition from design to development, enabling us to proceed with our project.

Day 2: Creating a Project Page in Flutter

Setting Up Flutter On the second day, we began creating a project page in Flutter. This involved:

- **Project Initialization**: Setting up a new Flutter project in Visual Studio Code.
- **Basic Structure**: Implementing the basic structure of the project page using the Flutter framework.
- **Testing Basics**: Applying the fundamental concepts learned from the Dart language to develop the initial page.

This task was essential for testing our understanding of Dart and Flutter, laying the groundwork for further development.

Day 3: Adding Project Files

Organizing Project Assets On the third day, we focused on organizing and adding necessary files within the project. This included:

- **Assets Folder**: Creating an assets folder to store images and fonts used in the project.
- **Screens Folder**: Setting up a screens folder to house all interface files, ensuring a clean and organized project structure.
- **Main File**: Keeping the main file separate to manage the primary application logic.

This organization facilitated easier management and integration of various components within the project.

Day 4: Identifying and Adding Libraries

Using pub.dev On the fourth day, we identified the libraries needed for our project using the pub.dev website. This process involved:

- **Library Research**: Finding libraries that could enhance our project's functionality and streamline development.
- **Dependency Management**: Adding the necessary libraries to the pubspec.yaml file.
- **Configuration**: Ensuring proper configuration and integration of the libraries within the project.

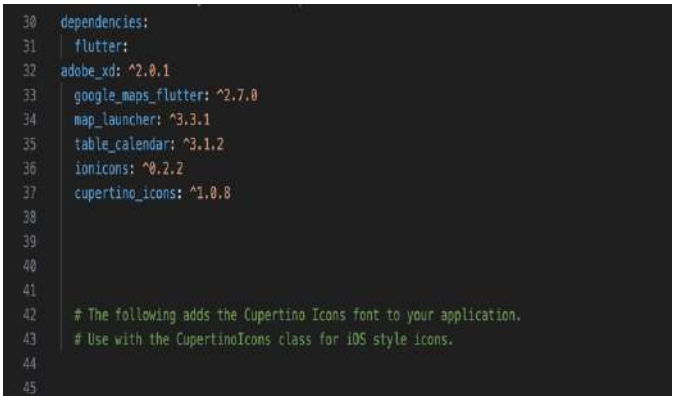
This step was crucial for leveraging existing tools and resources to improve our application's capabilities.

Day 5: Learning about Firebase

Overview of Firebase On the fifth day, we focused on learning about Firebase, a powerful platform for app development. We covered:

- **Real-Time Database**: Understanding how Firebase's real-time database can be used for storing and syncing data in real-time.
- **Cloud Messaging**: Exploring Firebase Cloud Messaging (FCM) for sending notifications and messages to users.
- **Authentication**: Learning about Firebase Authentication to implement secure user login and registration features.

This overview provided us with valuable knowledge on integrating Firebase into our project, enhancing its functionality and user experience.



```
dependencies:
  flutter:
adobe_xd: ^2.0.1
  google_maps_flutter: ^2.7.0
  map_launcher: ^3.3.1
  table_calendar: ^3.1.2
  ionicons: ^0.2.2
  cupertino_icons: ^1.0.8




  # The following adds the Cupertino Icons font to your application.
  # Use with the CupertinoIcons class for iOS style icons.
```

Figure 10.Add the necessary libraries.

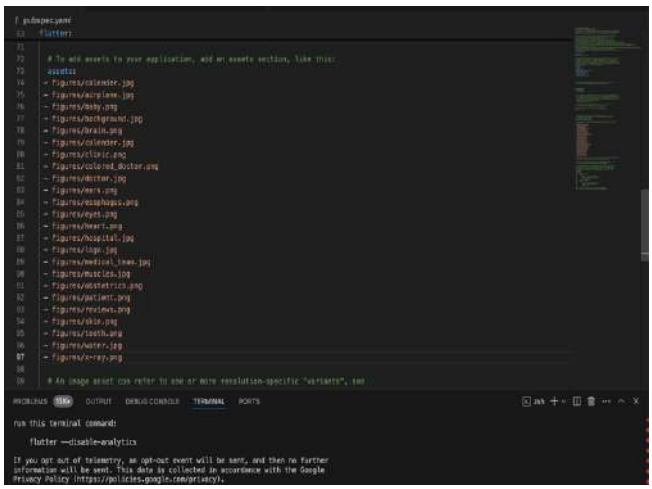


Figure 9.Add images to assets.

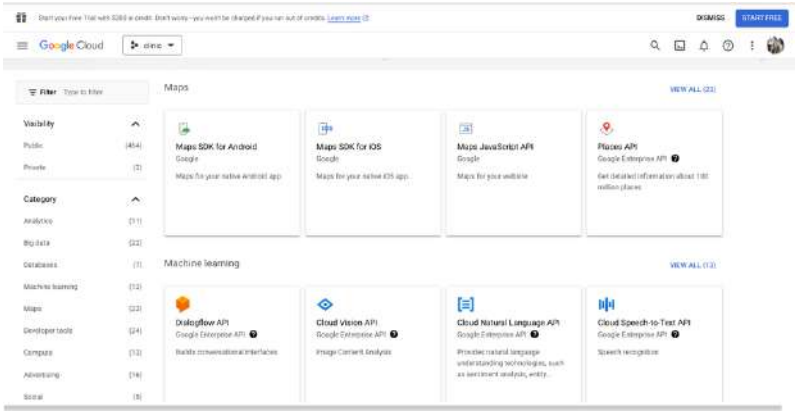


Figure 11.Google cloud library.

Figure 12.Realtime Database.

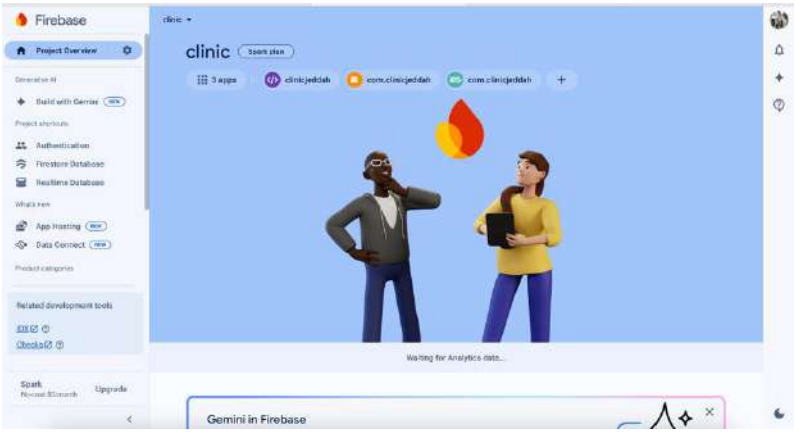


Figure 13.Create project in firebase.

Task Description: Fifth Week

Day 1: Exporting Code and Error Correction

Exporting from Adobe XD On the first day, we exported the design code from Adobe XD to Visual Studio Code. However, this process resulted in numerous errors within the code.

Correcting Code Errors To address these errors, we:

- **Analyzed Error Messages**: Identified and understood the specific issues causing the errors.
- **Referenced XD File**: Returned to the original XD file to cross-check and correct discrepancies.
- **Code Adjustments**: Made necessary adjustments and corrections in the code to align with the design specifications.

This initial troubleshooting was crucial for ensuring that our exported code was functional and accurate.

Day 2: Continued Error Correction

Ongoing Debugging On the second day, we continued to correct the errors identified in the code. This involved:

- **Detailed Debugging**: Systematically going through the code to fix remaining issues.
- **Testing Fixes**: Running the application to verify that the corrections resolved the errors.
- **Collaborative Problem Solving**: Working together to identify and address persistent problems efficiently.

This day was dedicated to ensuring that the code was stable and free from critical issues.

Day 3: Running the Application

Initial Run and Issues On the third day, we attempted to run the application, but encountered a significant problem: none of the application's pages were displaying correctly.

Diagnosis We diagnosed the issue by:

- **Reviewing Code Structure**: Checking the structure and connections between different components and pages.
- **Error Tracking**: Identifying the root cause of the missing pages through error messages and logs.

This step was vital for understanding and resolving the display issues within the application.

Day 4: Resolving Page Display Issues

Page Integration On the fourth day, we focused on solving the problem of missing pages. We ensured that the application correctly displayed the necessary pages, including:

- **Login Page**: Setting up the initial login interface.
- **Registration Page**: Implementing the registration functionality.
- **Main Page**: Configuring the main page of the application to be accessible after login.

This task was essential for establishing the user flow and ensuring a functional application structure.

Day 5: Adding Icons and Images

Enhancing UI with Assets On the fifth day, we added the necessary icons and images to the application. This involved:

- **Asset Integration**: Importing and incorporating images and icons into the project.
- **UI Enhancement**: Ensuring that all visual elements were properly displayed and aligned with the design.
- **Final Adjustments**: Making final tweaks to improve the overall look and feel of the application.

These additions were crucial for enhancing the user interface and ensuring a visually appealing application.

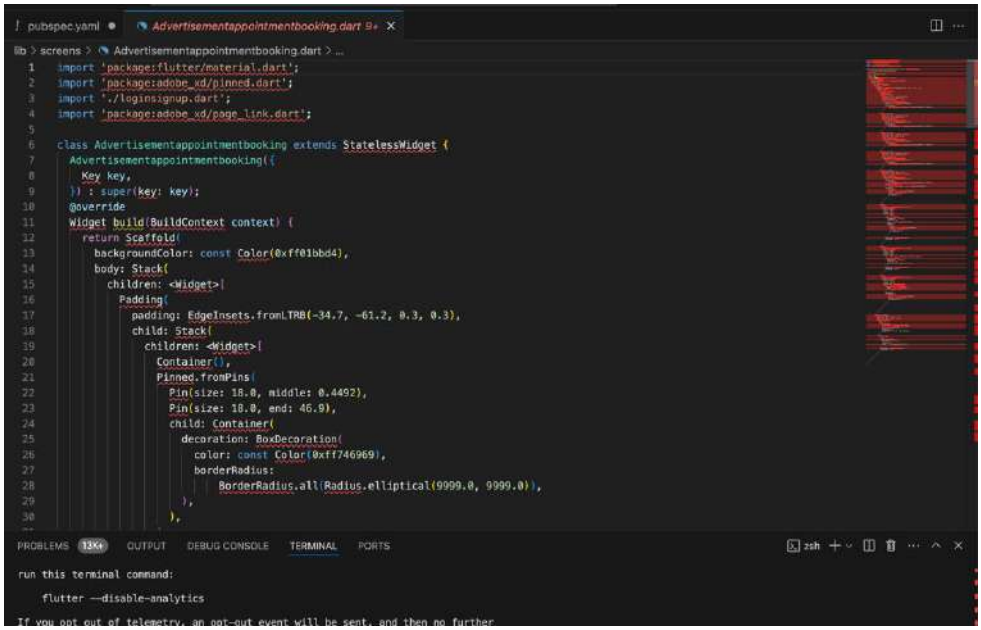


Figure 14.Errors after exporting the xd file to visual studio code.

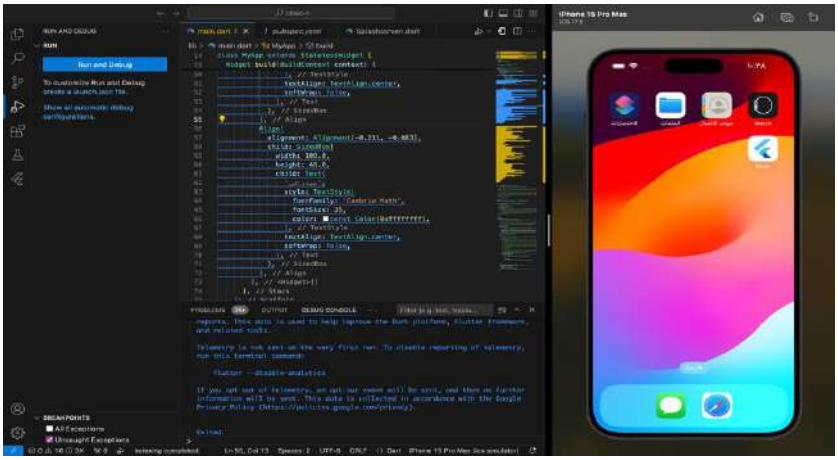


Figure 15.After running the code.

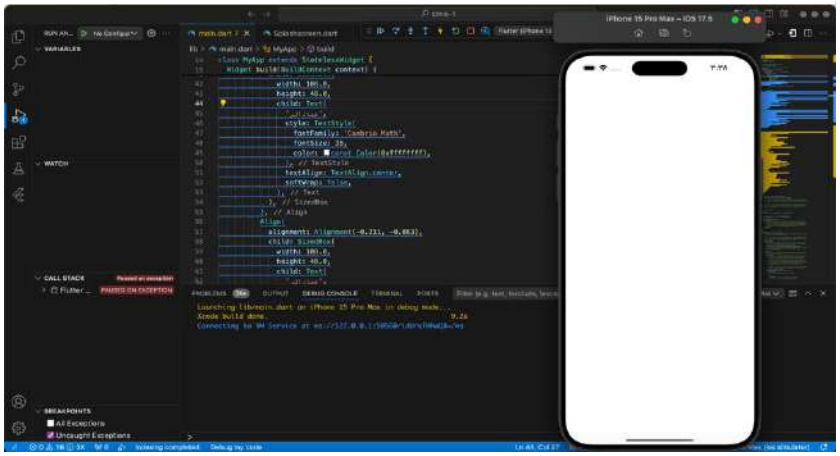


Figure 17.Pages not appearing after clicking the application.

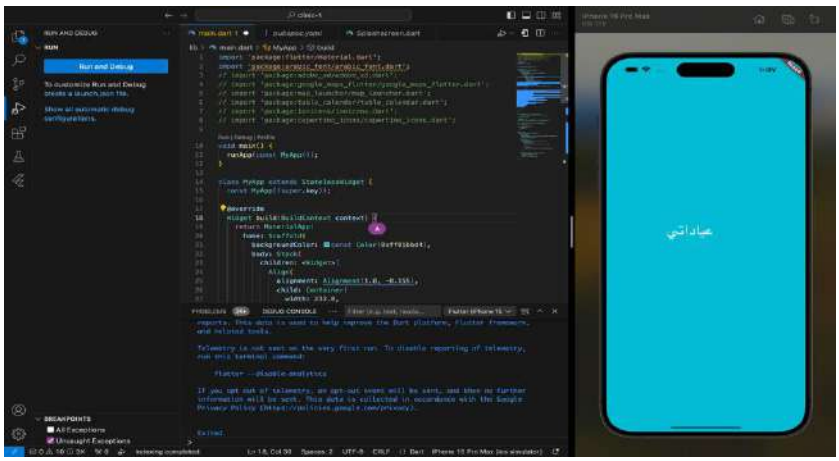


Figure 16.The first page of the application appears.

Figure 16. Registration page.

Figure 18.Login page.

Figure 19.Home page.

Figure 22.Doctors page.

Figure 21.Confirm appointment.

Figure 20.Clinics page.

Figure 25.Upcoming appointments page.

Figure 24.Old appointments page.

Figure 23.Evaluate page.

Conclusion

The summer training program at App Makers has been an enriching and transformative experience. Over five weeks, we progressed from initial design to developing a fully functional application, overcoming challenges and acquiring valuable skills along the way.

Key achievements include mastering tools like Visual Studio Code and Adobe XD, learning the Dart programming language, and integrating advanced features such as Firebase for real-time databases and authentication. Our teamwork and problem-solving skills were significantly enhanced through collaborative efforts and continuous learning.

We extend our heartfelt gratitude to App Makers and our supervisors for their guidance and support, which were crucial to our success. This training has not only equipped us with practical skills but has also inspired us to pursue further growth in app development.

In summary, this program has laid a strong foundation for our future careers, and we are excited to apply what we have learned to new projects and challenges.

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